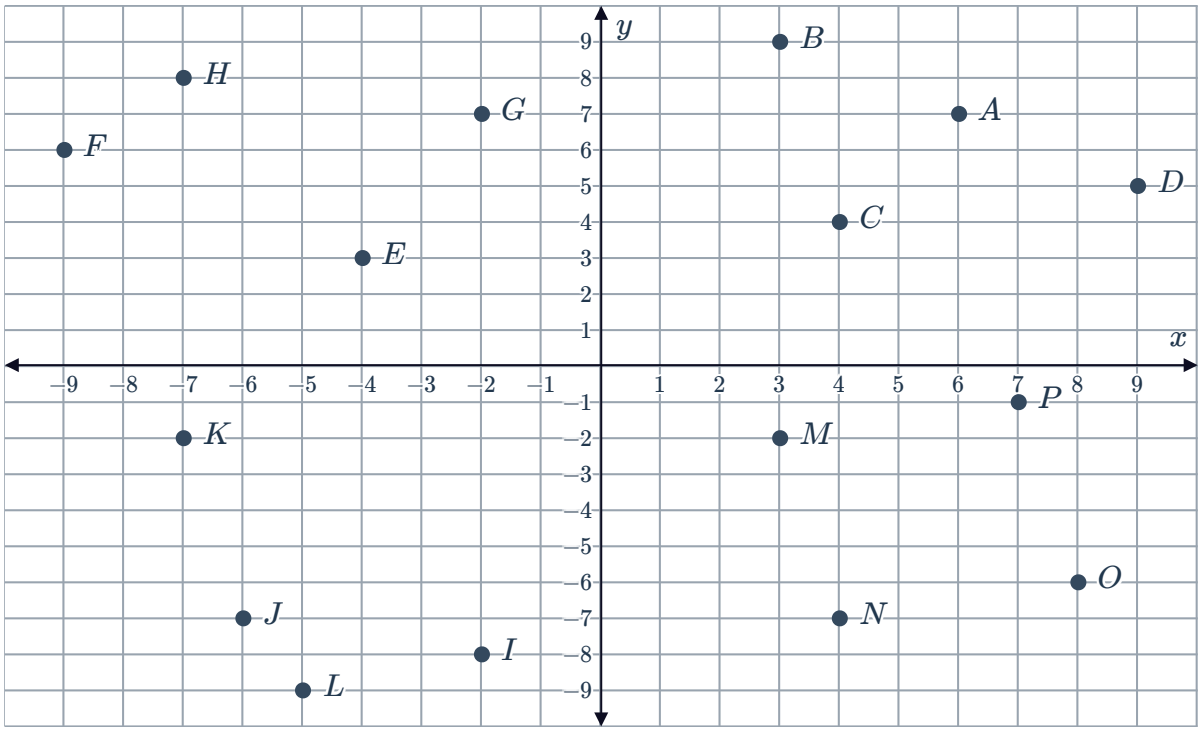
Directed numbers on the number plane

1 Consider the following number plane:



State the coordinates of the following points:

a A **b** B **c** C **d** D

e E **f** F **g** G **h** H

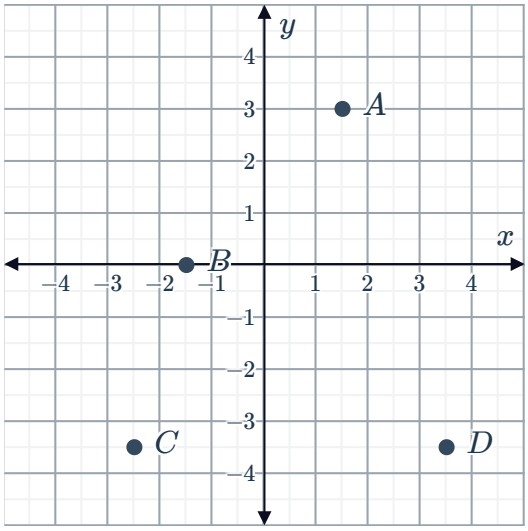
i I **j** J **k** K **l** L

m M **n** N **o** O **p** P

2 Consider the following number plane:

State the coordinates of the following points:

a A **b** B **c** C **d** D



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3

a State the x-coordinate of:

i The origin.

ii Any point on the y-axis.

b State the y-coordinate of:

i The origin.

ii Any point on the x-axis.

4 If point P has an x-coordinate of 0, which axis must it lie on?

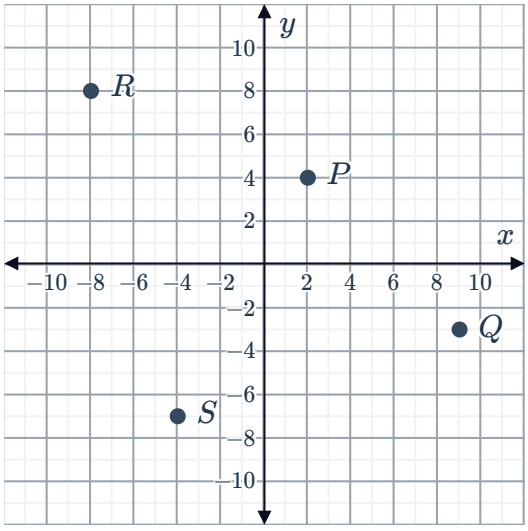
5 If (a, b) is a point that lies on the y-axis, find the value of a.

6 Consider the given points on the number plane:

State which point has the following coordinates:

a $(9, -3)$ **b** $(-4, -7)$

c $(2, 4)$ **d** $(-8, 8)$

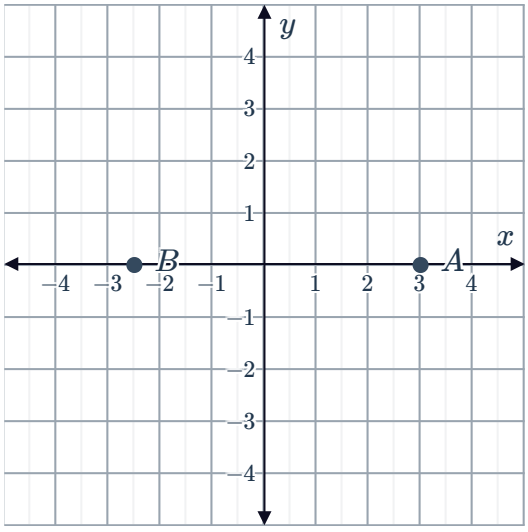


7 Consider the points A and B plotted on the number plane:

a Write down the coordinates of point A.

b Write down the coordinates of point B.

c Which axis do points A and B lie on?



8 In which quadrant do the following points lie?

a $(7, 8)$ **b** $(-3, 4)$ **c** $(2, -1)$ **d** $(-5, -6)$

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9 State the quadrant(s) that have:

a Points with a negative y-coordinate.

b Points with a negative x-coordinate.

c Points where the x-coordinate and y-coordinate have the same sign.

d Points with a negative x-coordinate and a positive y-coordinate.

e Points with a negative x-coordinate and a negative y-coordinate.

10 Plot the following points on a number plane:

a $A\,(6, 1)$ **b** $B\,(-7, 3)$ **c** $C\,(7, -4)$ **d** $D\,(-7, -3)$

e $E\,(-4, -4)$ **f** $F\,(-6, 0)$ **g** $G\,(2, 6)$ **h** $H\,(2, -8)$

i $I\,(-6, 3)$ **j** $J\,(-5, -5)$ **k** $K\,(-5, 9)$ **l** $L\,(5, -9)$

m $M\,(3, -3)$ **n** $N\,(0, 8)$ **o** $O\,(1, 3)$ **p** $P\,(-4, 5)$

q $Q\,(-2, -4)$ **r** $R\,(4, -3)$ **s** $S\,(1, 0)$ **t** $T\,(0, 2)$

11 For each of the following points:

i Plot the point on a number plane. **ii** State which quadrant the point lies in.

a $(2, 5)$ **b** $(-2, 1)$ **c** $(-3, -5)$ **d** $(8, -5)$

e $(-8, -1)$ **f** $(4, -7)$ **g** $(0, 2)$ **h** $(5, 0)$

12 For each of the following sets of points:

i Plot the points on a number plane.

ii Identify the shape formed by joining the points in order.

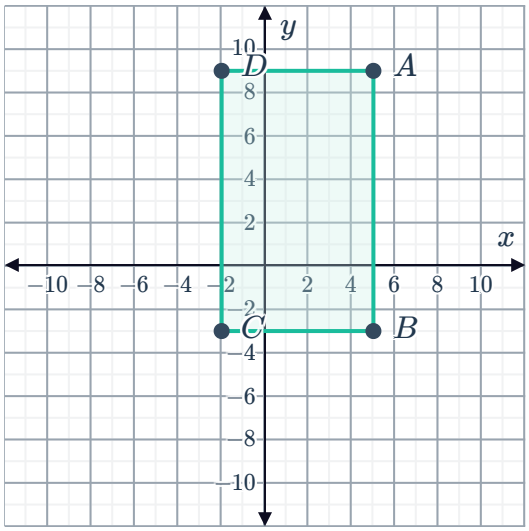
a $(4, 1), (7, 1), (4, 4), (7, 4)$ **b** $(1, 5), (6, 5), (1, 8), (6, 8)$

c $(-5, -3), (0, -1), (-4, 6), (1, 8)$ **d** $(-4, -3), (0, -1), (-2, 1), (2, 3)$

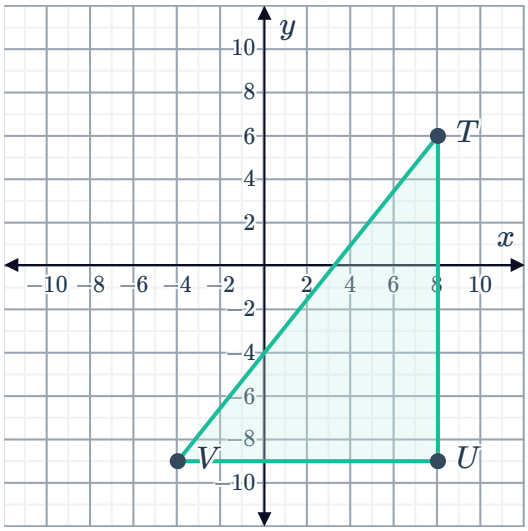
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13 For each of the following shapes, write down the coordinates of the vertices:

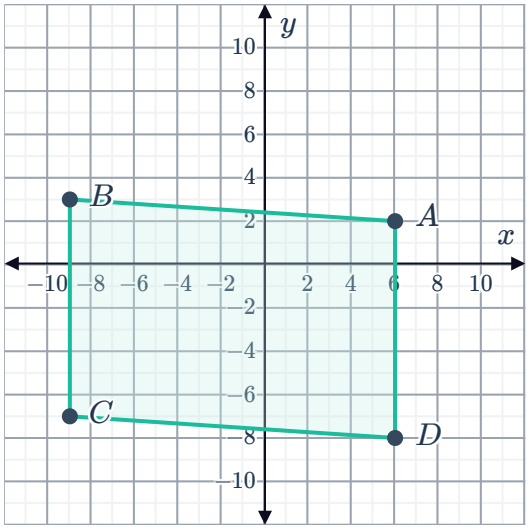
a



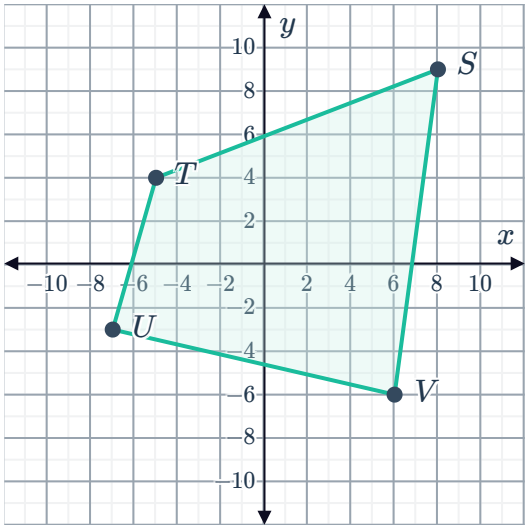
b



c



d



14 Describe how you would move on the number plane from the origin, to plot the following points:

a $(2, -4)$

b $(9, 14)$

15 Find the coordinates of the following points given the description:

a The point 9 units below the origin.

b The point 3.5 units to the left of the origin.

c The point 4 units to the left of $(-3, 6)$.

d The point 7 units to the right of $(-1, -2)$.

e The point 2 units to the right and 2 units below the point $(2, 5)$.

f The point 6 units to the left and 5 units above the point $(4, -4)$.

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16 Which point is furthest from the origin?

A $(0, -3)$ **B** $(5, 0)$ **C** $(0, 4.5)$ **D** $(-4, 0)$

17 How many units above the origin is the point $(-6, 1)$ located?

18 Starting at the origin, move 4 units left and then 4 units up.

a Plot the resulting point on a number plane.

b State the coordinates of this point.

Distance between two points

19 Find the distance between the following pairs of points:

a $A\,(-5, 8)$ and $B\,(-2, 8)$.

b $A\,(7, 3)$ and $B\,(-1, 3)$.

c $A\,(6, -5)$ and $B\,(6, -1)$.

d $A\,(-6, 2)$ and $B\,(-6, -7)$.

20 Consider the points: $A\,(-4, 8)$, $B\,(-7, 8)$ and $C\,(-7, 1)$.

a Plot the points on a number plane.

b Find the length of AB.

c Find the length of BC.

21 A triangle has points $A\,(1, 2)$, $B\,(-2, -3)$ and $C\,(6, -3)$.

a Plot the triangle ABC on a number plane.

b Find the perpendicular height of the triangle if BC is the base.

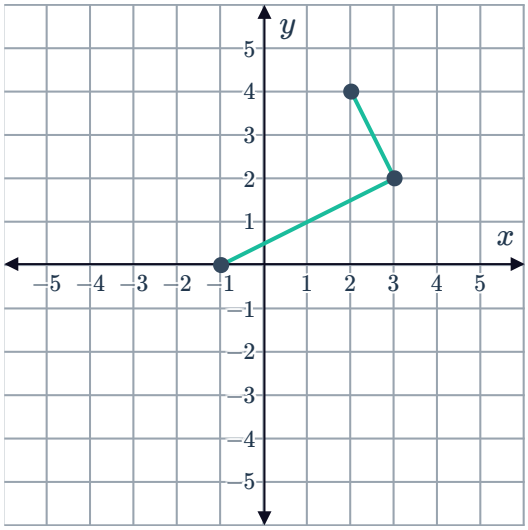
c Find the length of base BC.

d Find the area of triangle ABC.

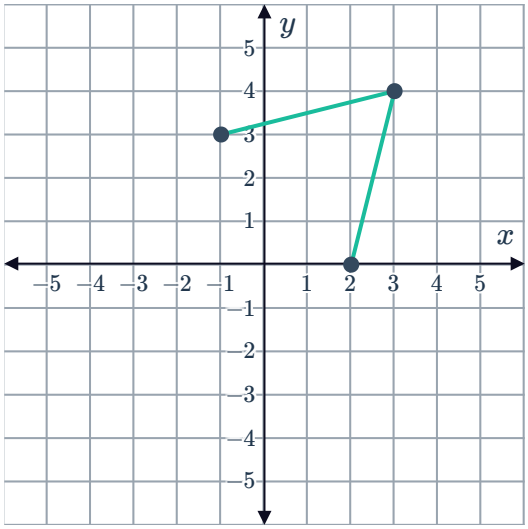
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Geometric applications

22 The points given represent three vertices of a parallelogram. Find the coordinates of the fourth vertex if it is known to be in the 2nd quadrant.



23 The points given represent three vertices of a rhombus. Find the coordinates of the fourth vertex if the missing point lies in the 3rd quadrant.



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